

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
6				Indicative content Boiling - boiling causes calcium hydrogencarbonate to decompose (or magnesium hydrogencarbonate) - a precipitate of calcium carbonate is formed (limescale) - calcium / magnesium ions are removed from the water $\text{Ca}(\text{HCO}_3)_2 \rightarrow \text{CaCO}_3 + \text{H}_2\text{O} + \text{CO}_2$ - very simple method of softening - but limescale is formed which causes ‘furring’ / reduces boiler efficiency / blocks pipes - boiling only removes temporary hardness - not for large-scale as it requires a lot of energy references to ‘scum’ are neutral Ion exchange - ion exchange columns contain a resin (in small balls) coated with sodium ions - as hard water passes through the column, sodium ions come off the resin and go into the water, while calcium / magnesium ions come out of the water and stick to the resin - removes both temporary and permanent hardness - continuous process - resin needs changing / regeneration - water contains high levels of sodium ions which can be harmful for some people	6			6		3

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6				5-6 marks Good description of both methods with an advantage and/or disadvantage for each <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i> 3-4 marks Basic description of both methods; reference to an advantage and a disadvantage <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1-2 marks Some knowledge of one method and/or its advantage/disadvantage <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
				Question 6 total	6	0	0	6	0	3